Remote sensing large wood storage downstream of reservoirs during and after dam removal: Elwha River, WA, USA.

Manuscript figures

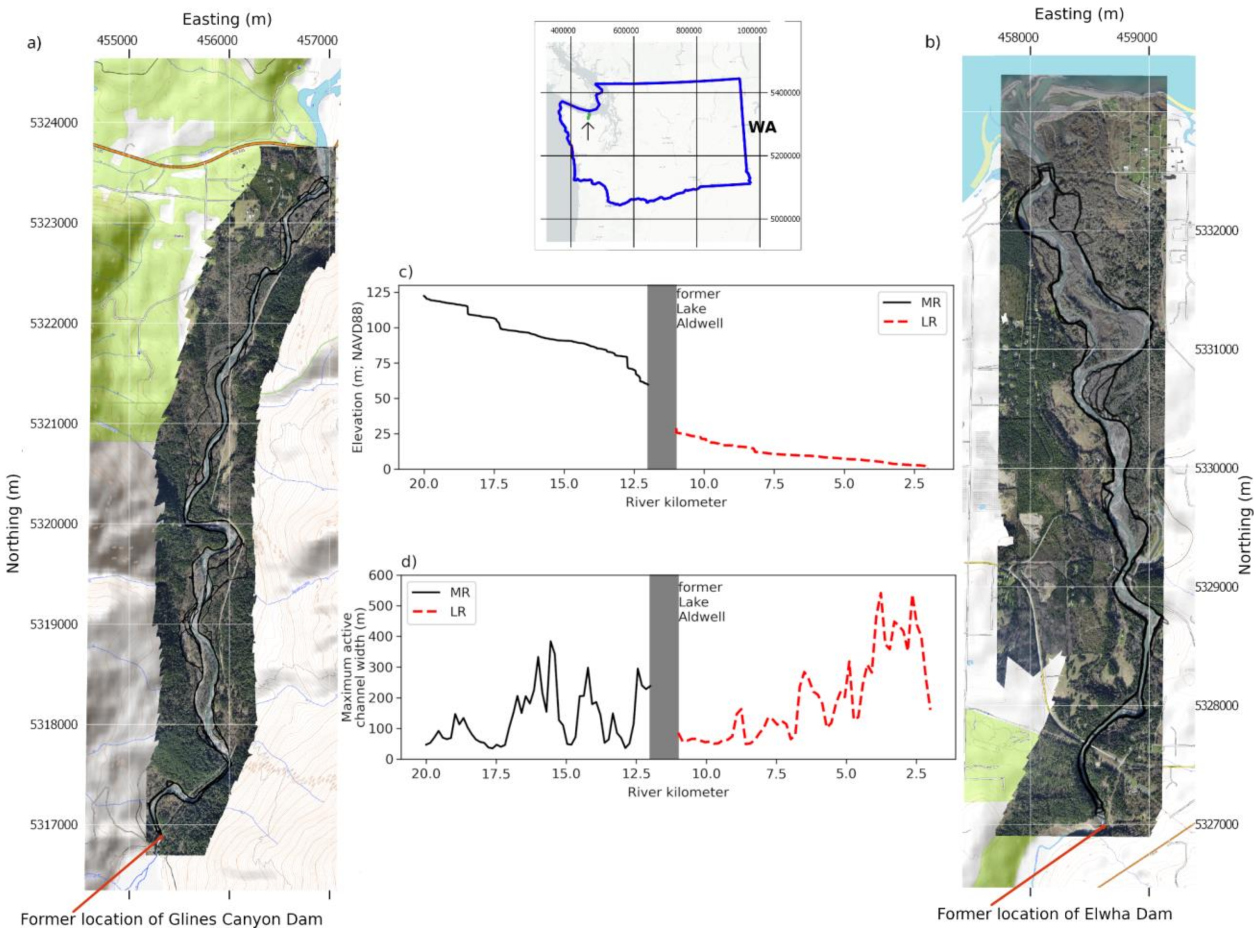

Figure 1: The two reaches of the Elwha River under consideration, namely a) the Middle Reach (MR) and b) the Lower Reach (LR), located in the Olympic peninsula of Washington state, USA (inset map) and draining into the Strait of Juan de Fuca in the Salish Sea. The two reaches are illustrated using orthoimagery reconstructed from one aerial flight conducted on 2015-03-03. Imagery comes from Ritchie et al. (2018). The digitized active channel margins are from East et al. (2018). Eastings and Northings are NAD83(2011) UTM 10N. c) The elevation profile of the river. d) The maximum active channel width, determined from the 14-point time-series of orthoimagery that spans the study period.

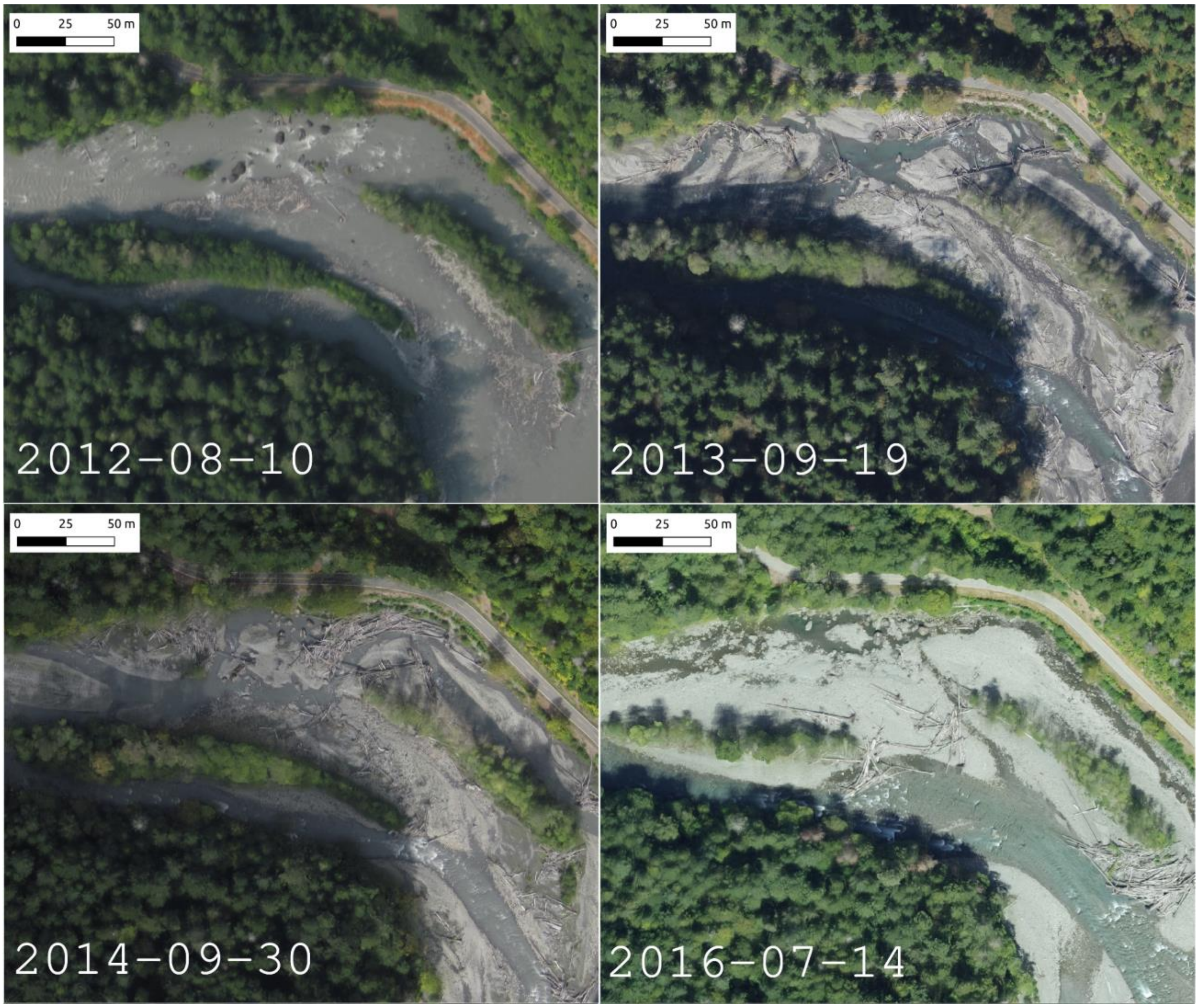

Figure 2: Example orthoimagery at a typical location (located in the middle reach), showing the variation in color and brightness, as well as the nature of the channel, wood deposition, and bar growth. Please see Supplemental Figure S1 for the full 14-image time-series of orthoimagery at this location, as well as the variation in average image brightness.

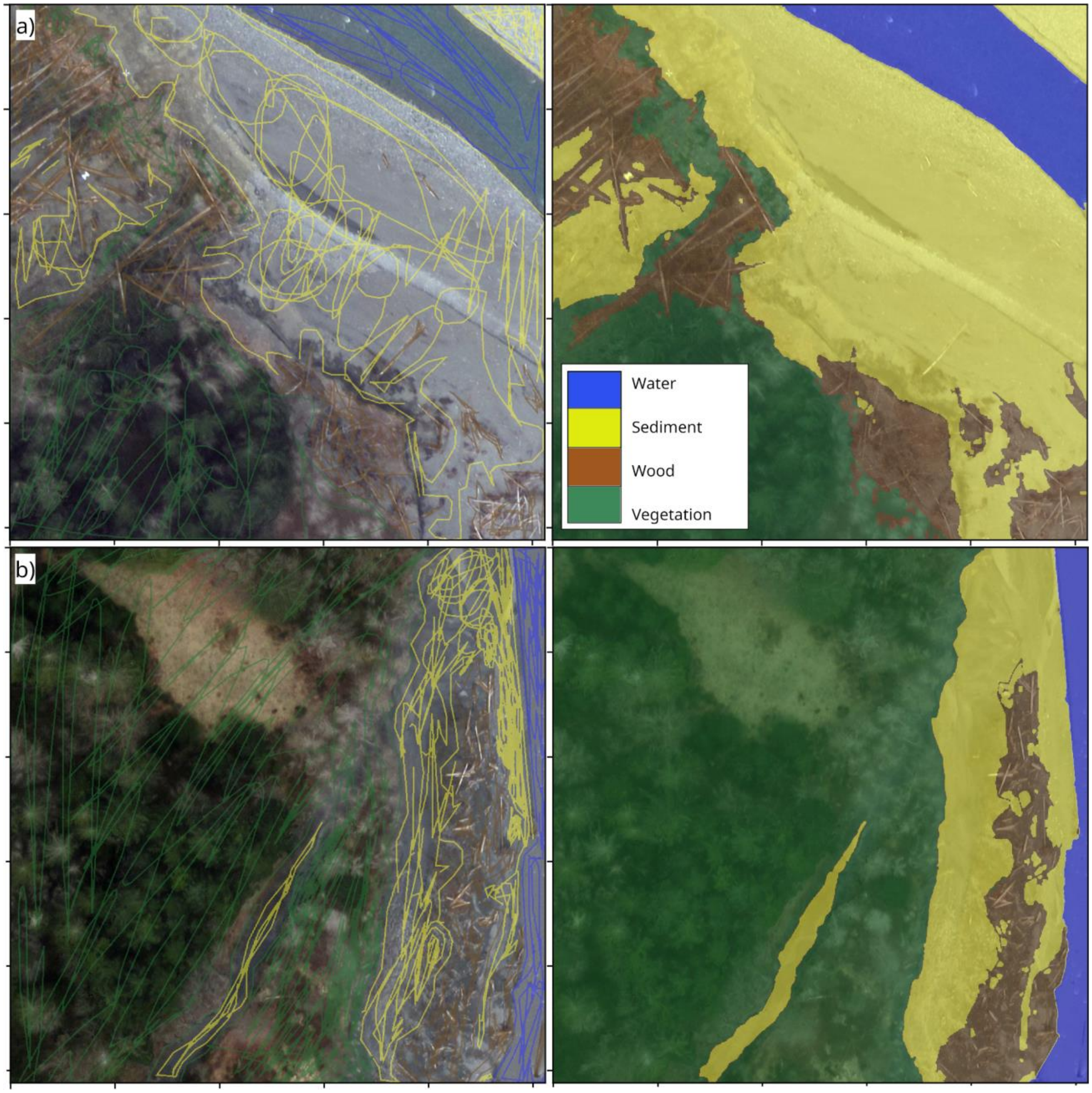

Figure 3: Example annotated images (left column) and resulting label images (right column) made using the software program, Doodler (Buscombe et al. 2021). Spacing between ticks is 200 pixels. Please see Supplemental Figure S14 for more examples of annotated imagery and labels.

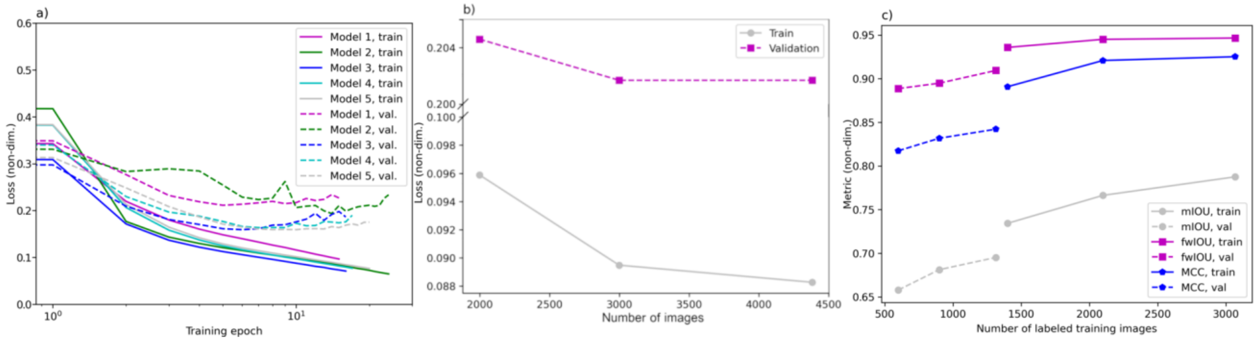

Figure 4: a) Loss curves for additional training runs for the 4-class image segmentation model (b) The minimum train and validation loss from the best four-class model trained with increasing numbers of labeled images. (c) Various train/validation metrics (mean IoU, frequency weighted IoU, and Matthews Correlation Coefficient) as a function of number of training/validation images.

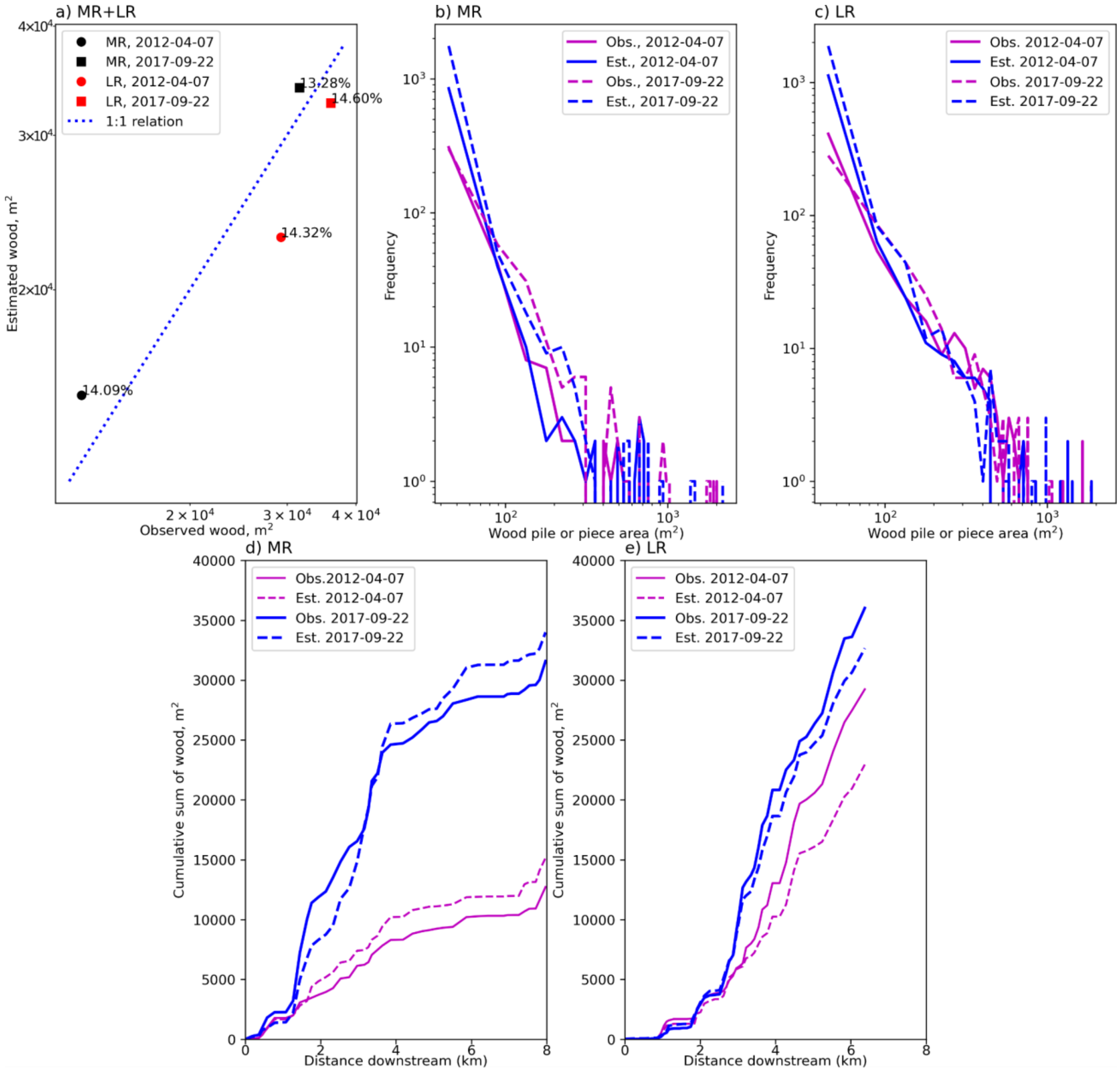

Figure 5: Model wood detection evaluation. Each panel shows results from all four sets of ground truth observations (from lower reach or LR and middle reach or MR, 2012-04-07 and 2017-09-22 respectively). a) Observed versus estimated total wood area; b) Observed and estimated wood deposit size-distributions in the MR; c) Observed and estimated wood deposit size-distributions in the LR; d) Cumulative sum of wood as a function of distance downstream in the MR; and e) Cumulative sum of wood as a function of distance downstream in the LR.

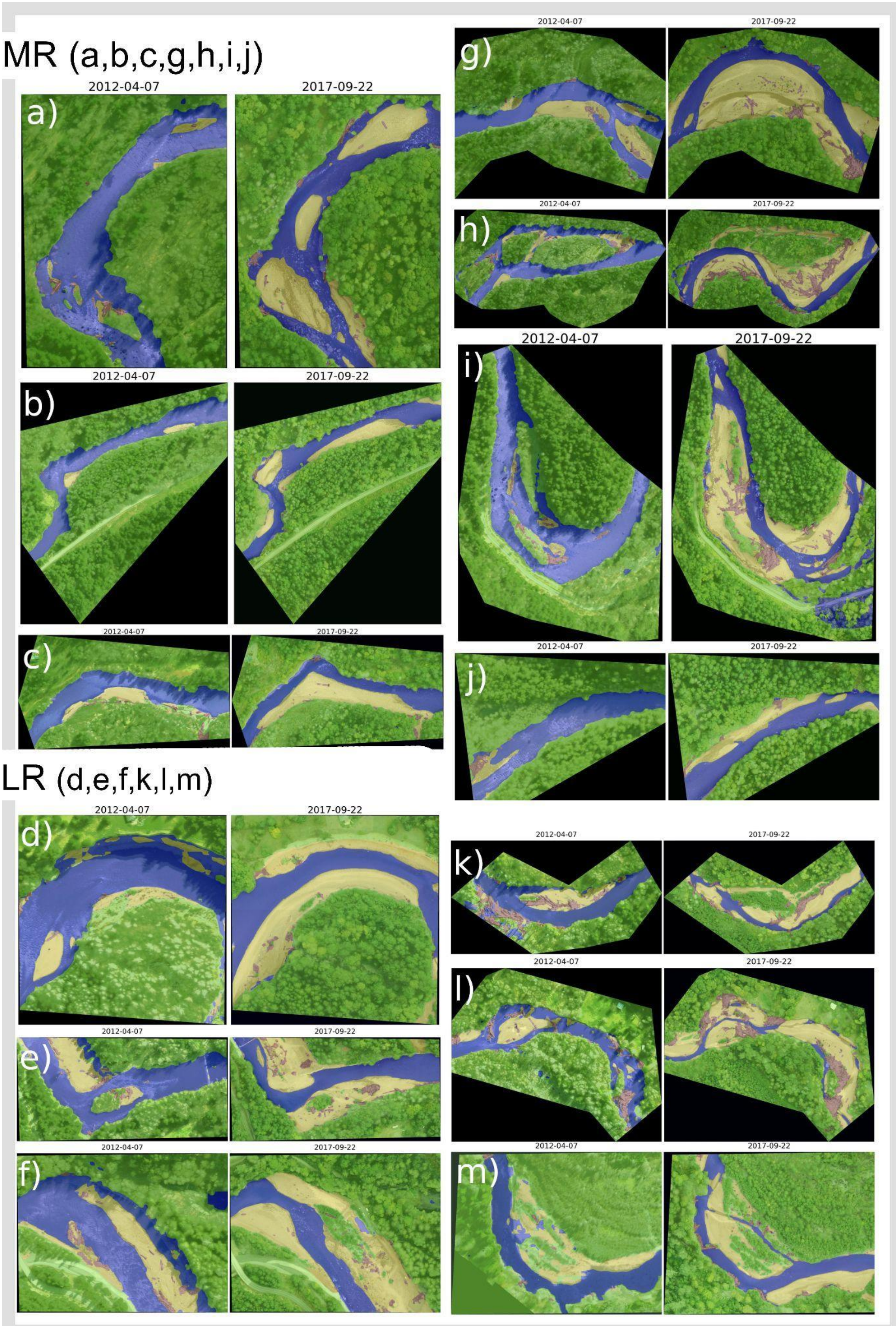

Figure 6: Example model outputs for the first (2012-04-07) and last (2017-09-22) from a selection of bars in the middle reach (MR) and lower reach (LR).

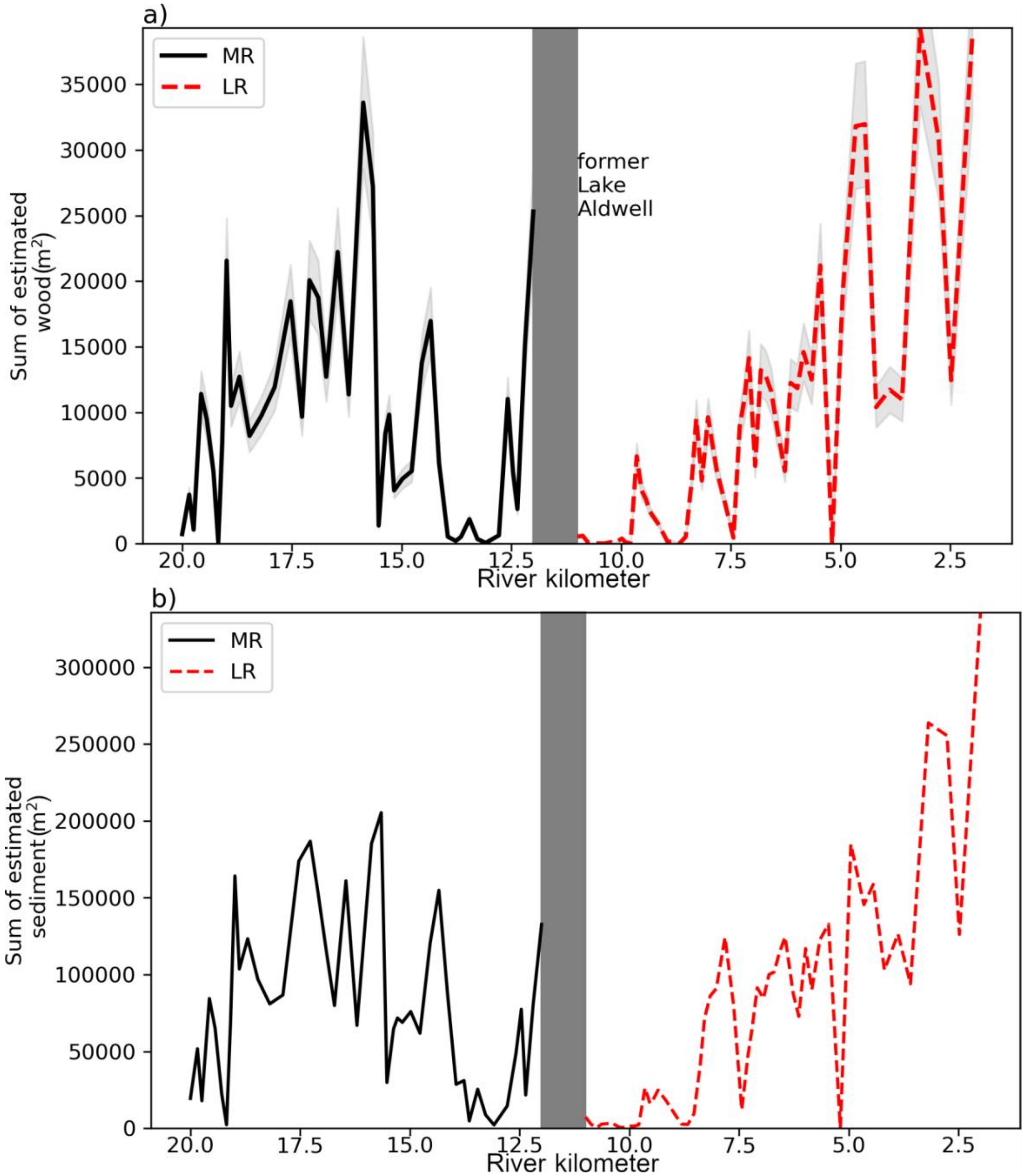

Figure 7: Total wood (a) and sediment (b) area as a function of downstream distance. Shading in a) is +/- 15%.

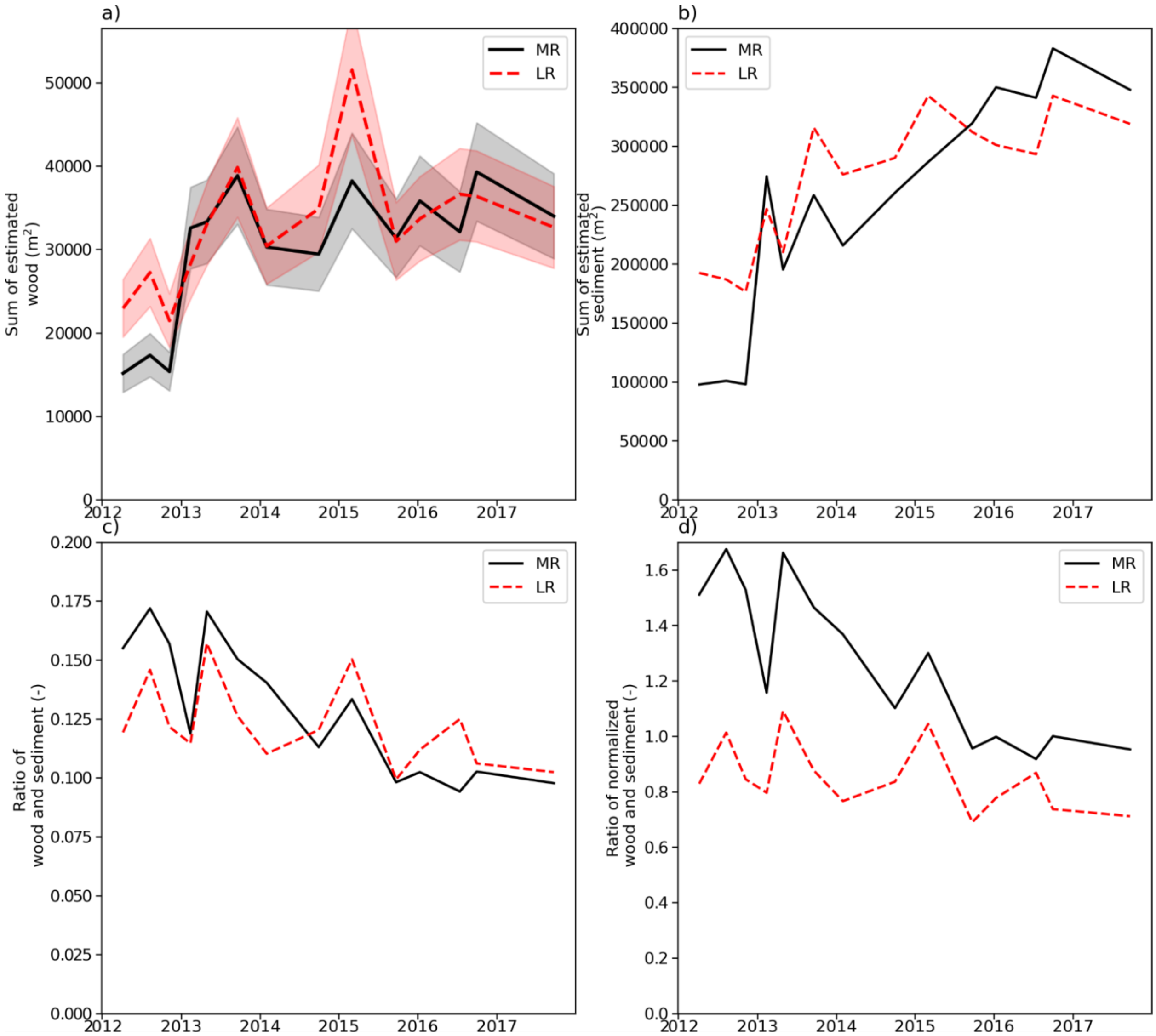

Figure 8: Time-series of total a) wood and b) sediment areas, c) ratio of wood and sediment areas, and d) normalized ratio of wood and sediment, computed as the normalized sum of wood area divided by the normalized sum of sediment area. Shading in a) is +/- 15%.

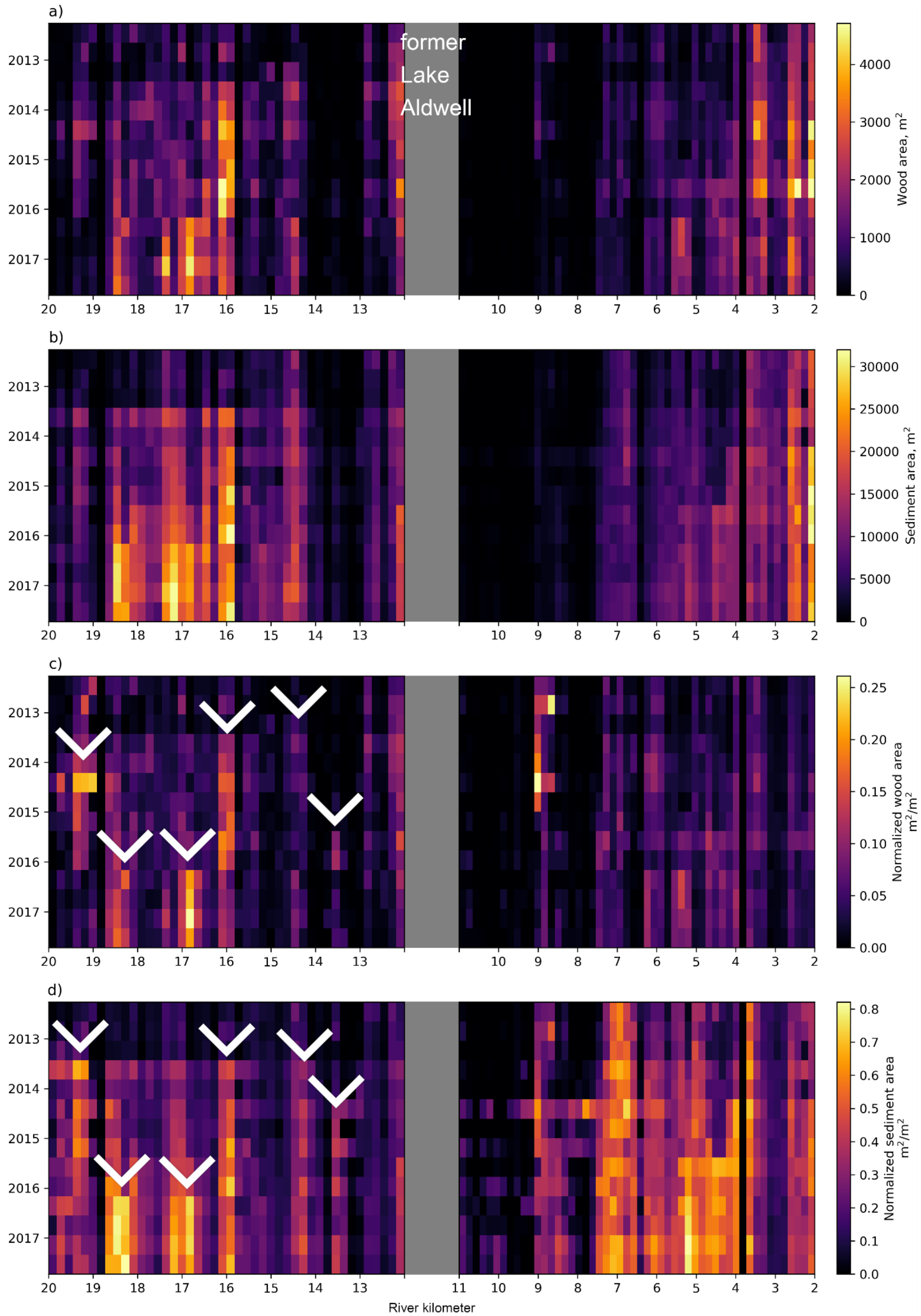

Figure 9: Wood and sediment deposit area (m^2) and normalized wood areal (i.e., wood concentrations) (m^2/m^2) in the middle reach (left column) and lower reach (right column). In each subplot, magnitudes are shown as a function of time (decreasing on the y axis) and distance downstream (decreasing on the x axis). Arrows in e) and g) refer to specific localized events discussed in the accompanying manuscript.

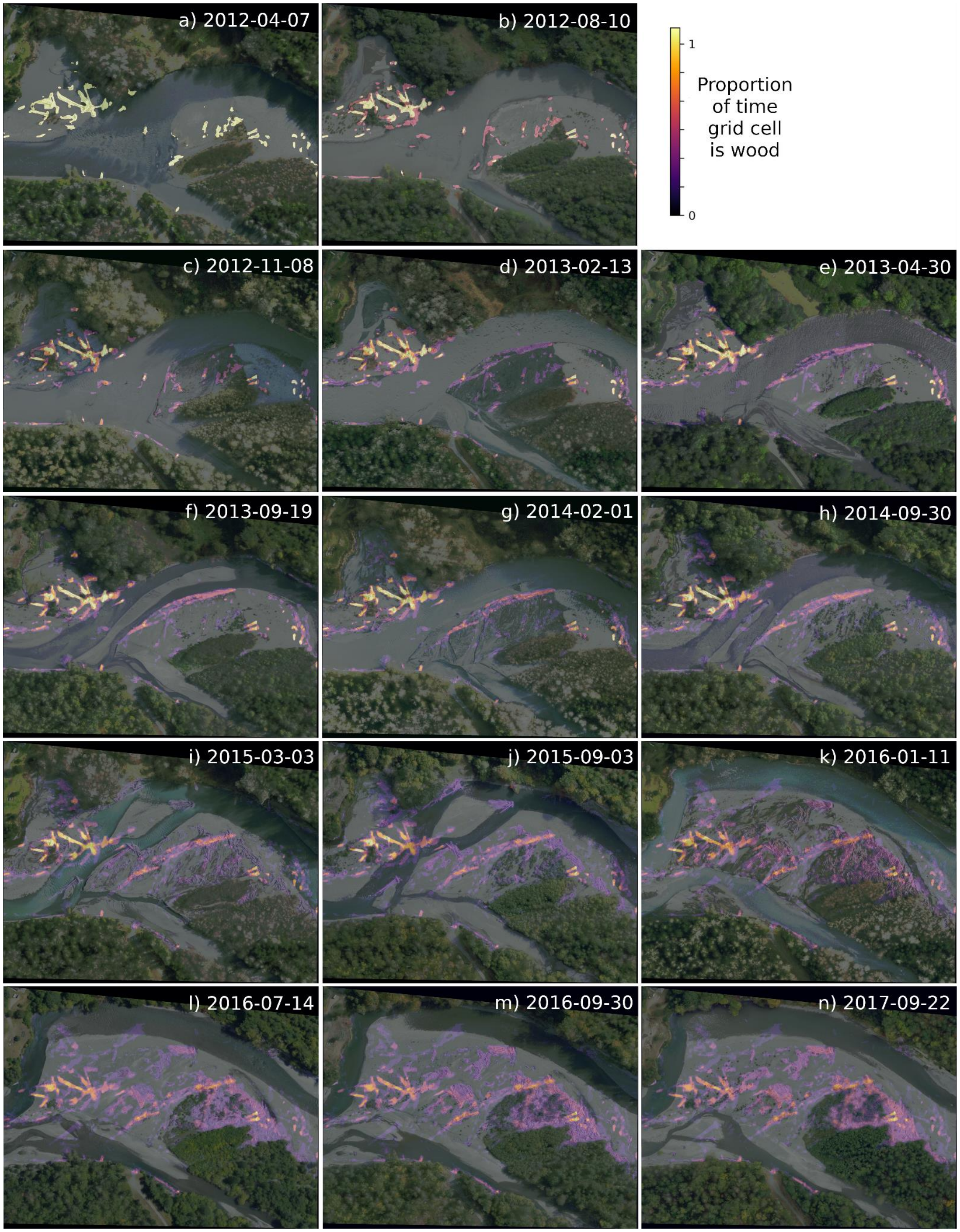

Figure 10: Large wood persistence in an example location within the middle portion of the lower reach in which there was modest amounts of pre-existing wood. Each panel shows an orthoimage at a different time in the 14-point time-series. The brightness of the superimposed persistence map is proportional to the proportion of cumulative time wood has been present at that location.

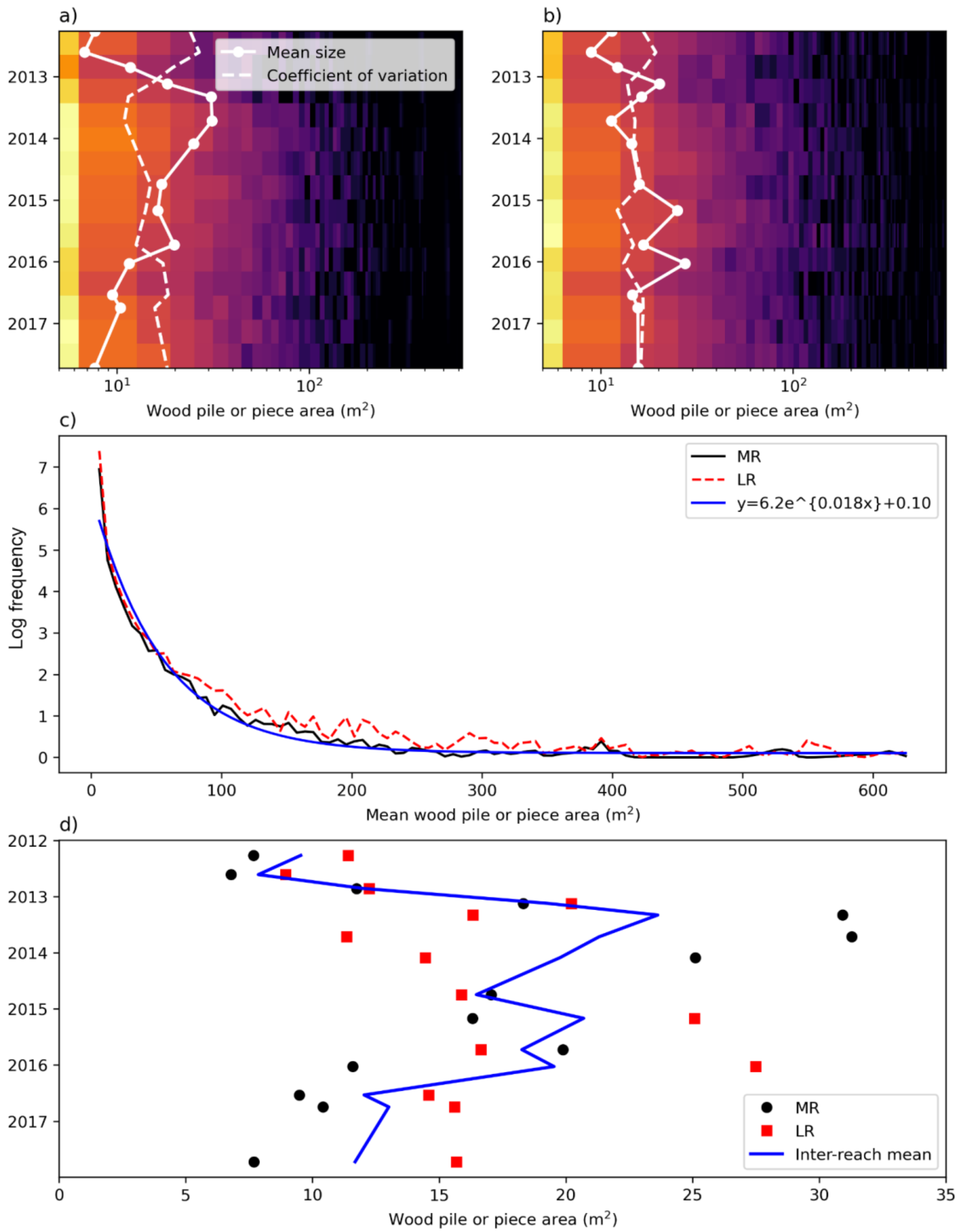

Figure 11: Time-varying wood size-distributions in the middle reach (MR; a) and lower reach (LR; b). The time-series of mean wood piece/pile size and coefficient of variation of wood sizes are also shown. c) The time-averaged wood size-distribution in both reaches, conforming to an exponential distribution model fit to the data obtained using optimization methods. d) Time-series of mean wood piece/pile size, and the inter-reach mean, which initially increased until 2014/2015, then decreased.

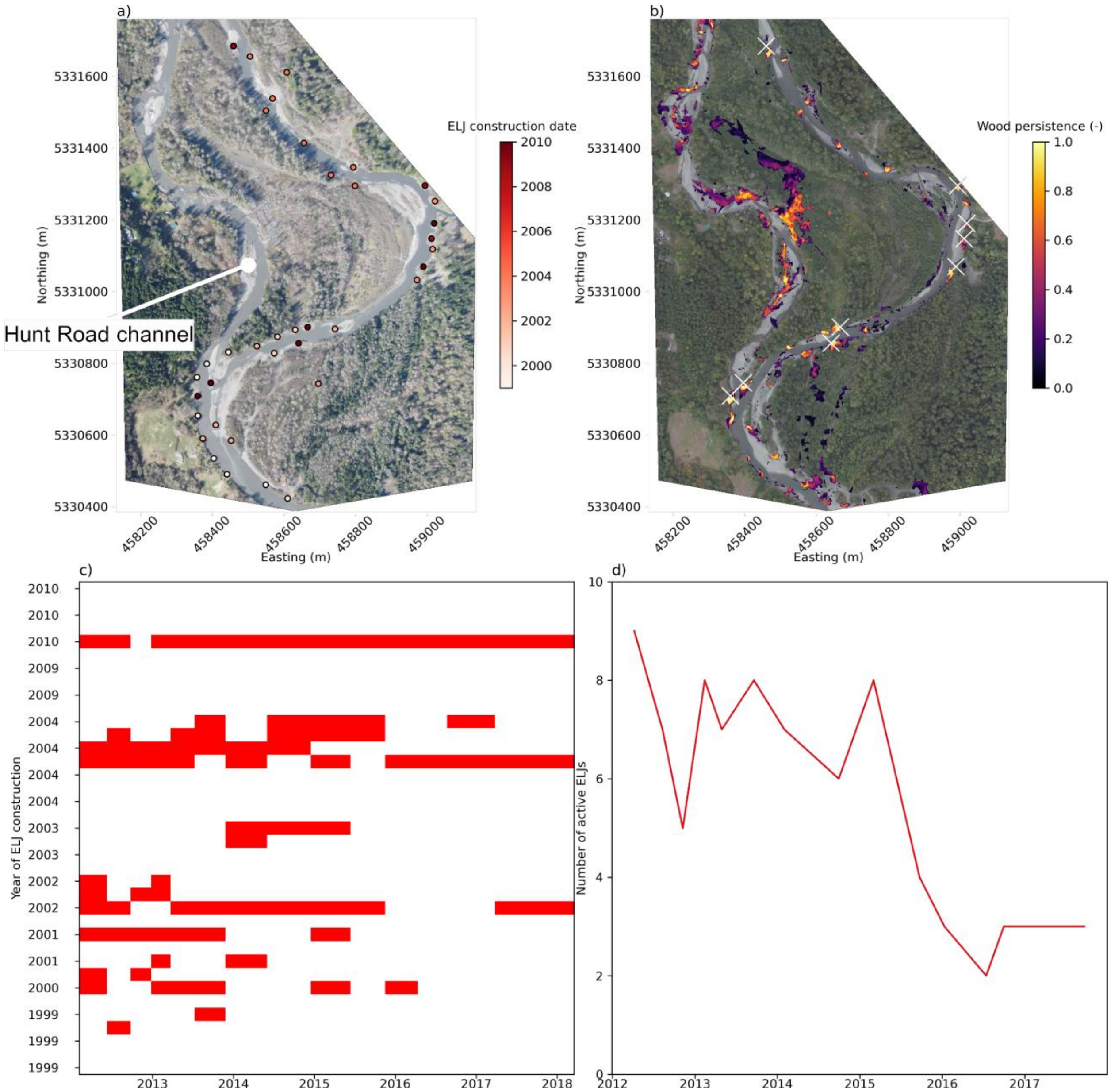

Figure 12: a) Locations of Engineered Log Jams (ELJs), colored by their date of construction (no new ELJs were constructed between 2010 and the end of the present study, September 2017.) Background image is from 2012-04-07. The river flows to the north, and the Hunt Road channel, which avulsed prior to this study, is indicated. b) The persistence of wood (the sum of wood pixels across all surveys) overlying the 2017-09-22 image, and with the post-2009 ELJ locations as white crosses. Note that the 2017-09-22 imagery in b) has had a gamma correction applied (with an exponent of 2.2) to enhance the brightness for better visibility. c) The presence (blue bar) and absence (absence of blue bar) at each ELJ (y axis), ordered by date of ELJ construction, and survey time (x axis). d) The number of active ELJs over time. Active ELJs are those with wood accumulations within 5m at time of survey.

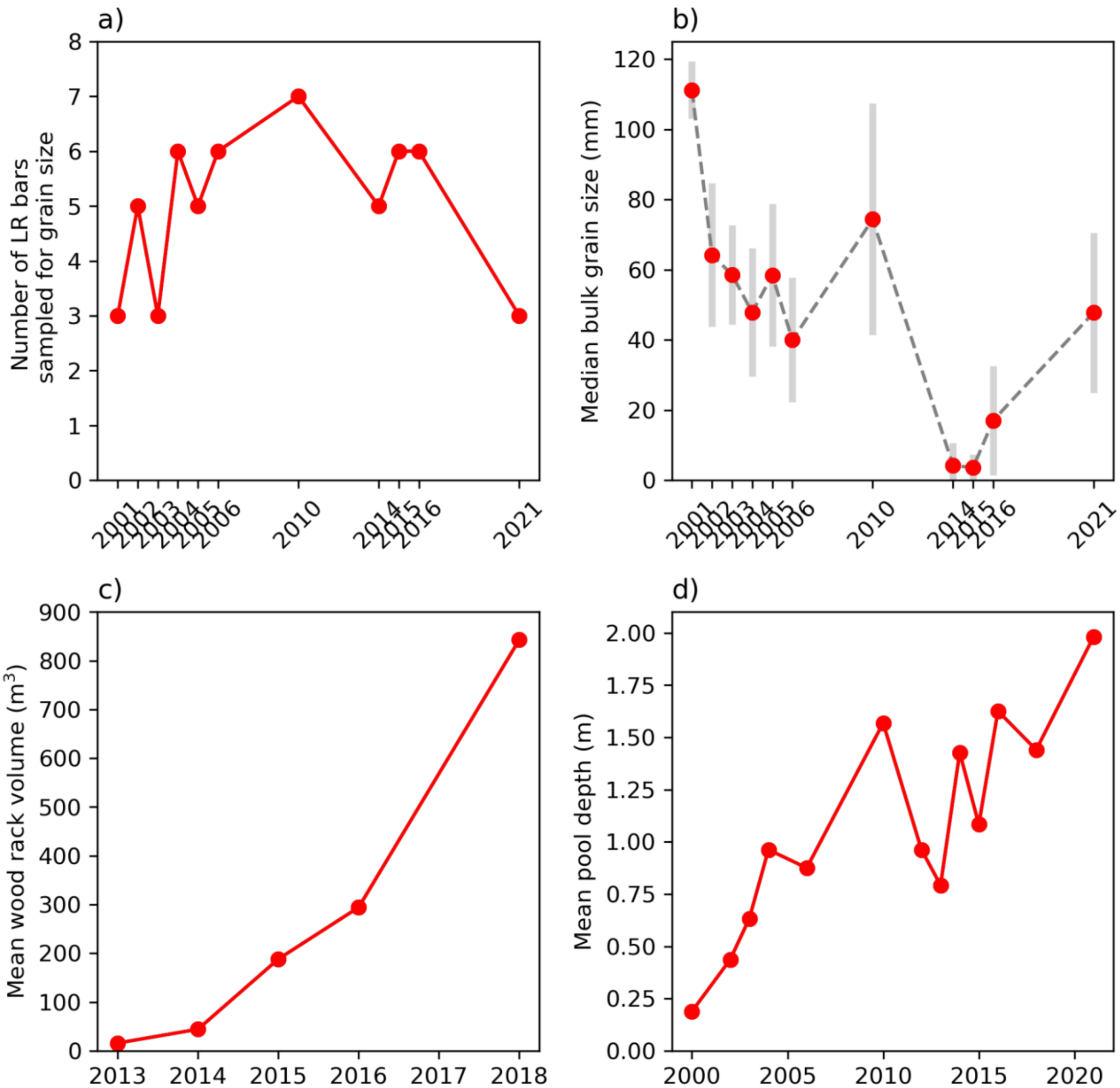

Figure 13: Time-series measurements of lower reach bar grain size, wood rack volumes, and pool depths. a) number of bars sampled for grain size; b) Median grain size of lower reach alluvial bars (gray bars indicate +/- 1 standard deviation); c) Mean wood rack volume; and d) Mean pool depth.

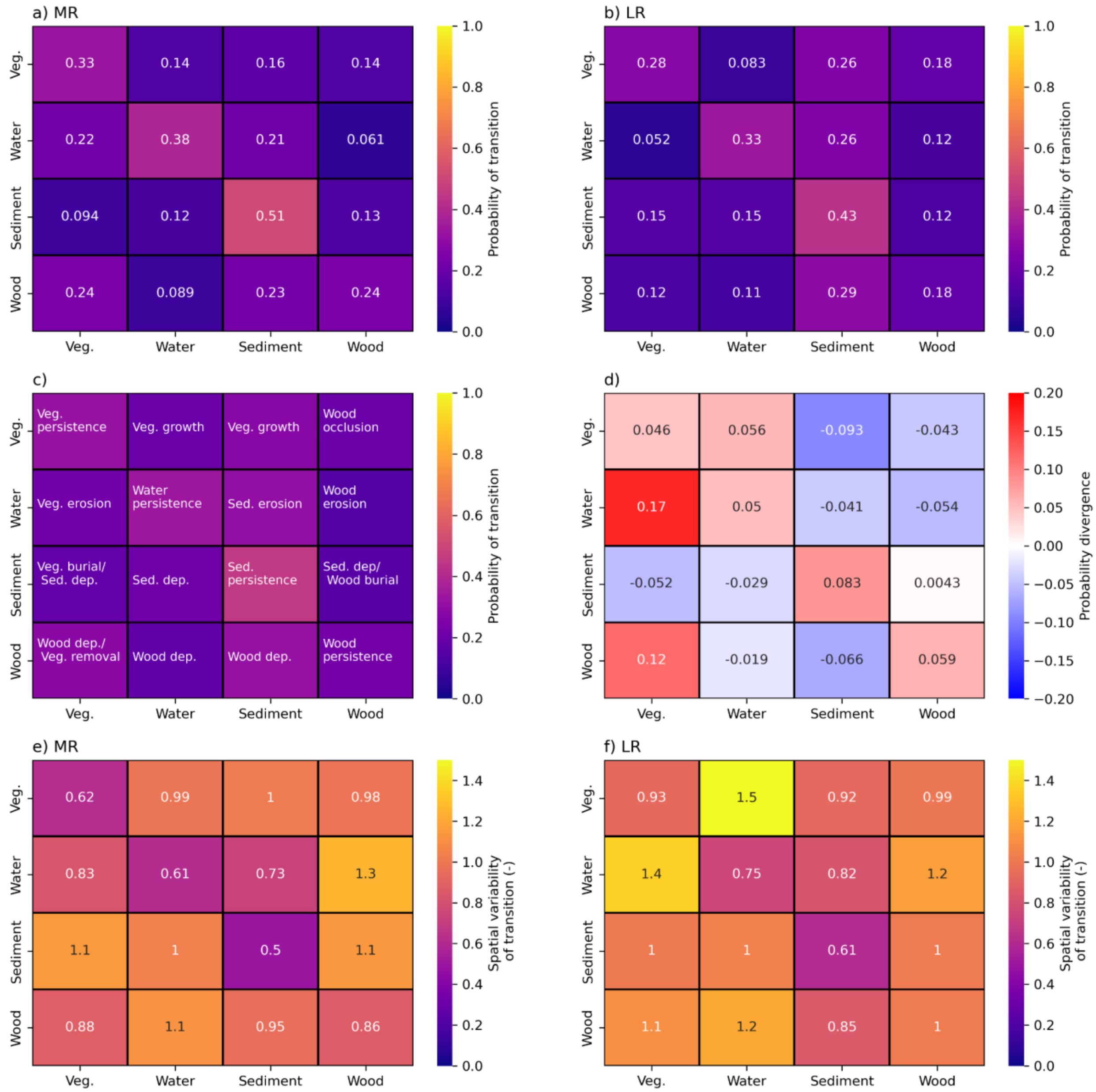

Figure 14: Transition probability matrices for the a) middle reach (MR) and b) lower reach (LR), based on an analysis of all 14 four-class (vegetation, water, sediment, wood) label maps. c) A schematic of the proposed 16 processes quantified by a transition matrix. d) Divergence in probabilities between reaches, computed as MR minus LR, therefore positive numbers mean the process is more prevalent in the MR, and negative numbers indicate a process more prevalent in the LR. e) Coefficient of variation of TPMs computed over all subreaches of the MR and f) LR. The coefficient of variation is computed per cell as the standard deviation in transition probabilities divided by the mean transition probabilities. Large values therefore quantify highly spatially episodic (fluctuating) processes, whereas small values indicate spatial consistency.